



Project: Design a full-fledged network for an organization with multiple subnets.

Course: CSE 405

Section: 03

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Introduction

Apex University, is an enterprise like East West University, owns many computers, with a complex network infrastructure. Apart from wired internet access to all the classrooms, labs, employee PCs, library and other administrative and academic wings, the university also provides wireless internet access for every campus. On top of that the university runs complex networked systems to support several of its business process like admissions, advising, results, eTender, library management, accounts and so on.

This complex network infrastructure is subnetted and switching/routing mechanisms are in practice.

Tools

1. CISCO Packet Tracer 8.2.2
2. PC-PT, Laptop-PT, SmartPhone-PT
3. Router (Generic Router-PT)
4. Wireless (AccessPoint-PT)
5. Switch (2960 IO S 15)
6. Server (DNS Server, DHCP Server, Web Server)
7. Connector (Copper Straight-Through, Serial DCE)

Tasks

1. The web page of the university reflecting its own information.
2. A single DNS sever is installed to locate web server - meaning people will browse University's web site from any campus with the following address:
<http://www.apex.edu.bd>
3. Configured the whole network in such a way that IP for the hosts of different campuses is automatically assigned by a single DHCP server.
4. Created five LAN network, one for each campus; among the hosts in a network making sure some wireless hosts are added in addition to wired hosts.
5. University's full network has covered its five campuses with five routers; Connections between the campus routers are established strictly following the given topology.
6. Connectivity between all the hosts are established.
7. For routing protocol, dynamic routing algorithm i.e., OSPF is used.
8. IP addresses are assigned as instructed: Campus 1 and Campus 2 Network addresses are from "A" class; Campus 3 and Campus 4 Network addresses are from "B" class and Campus 5 is from "C" class. Placement of all 3 servers i.e., WEB server, DNS server and DHCP server are in campus 5.
9. Connection between all the routers are in serial connections.
10. Further expansion and growth are possible if need be.

Network Layout

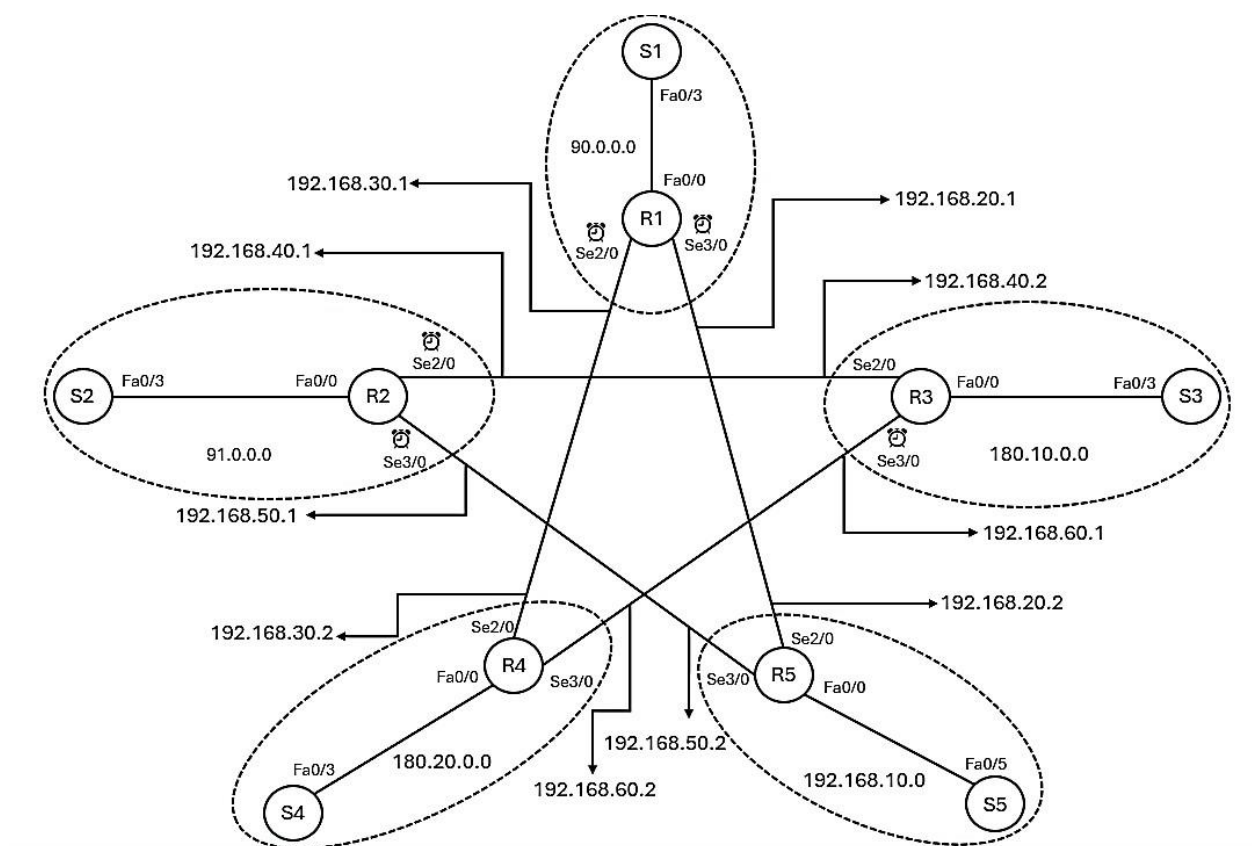


Fig-1

Network Diagram

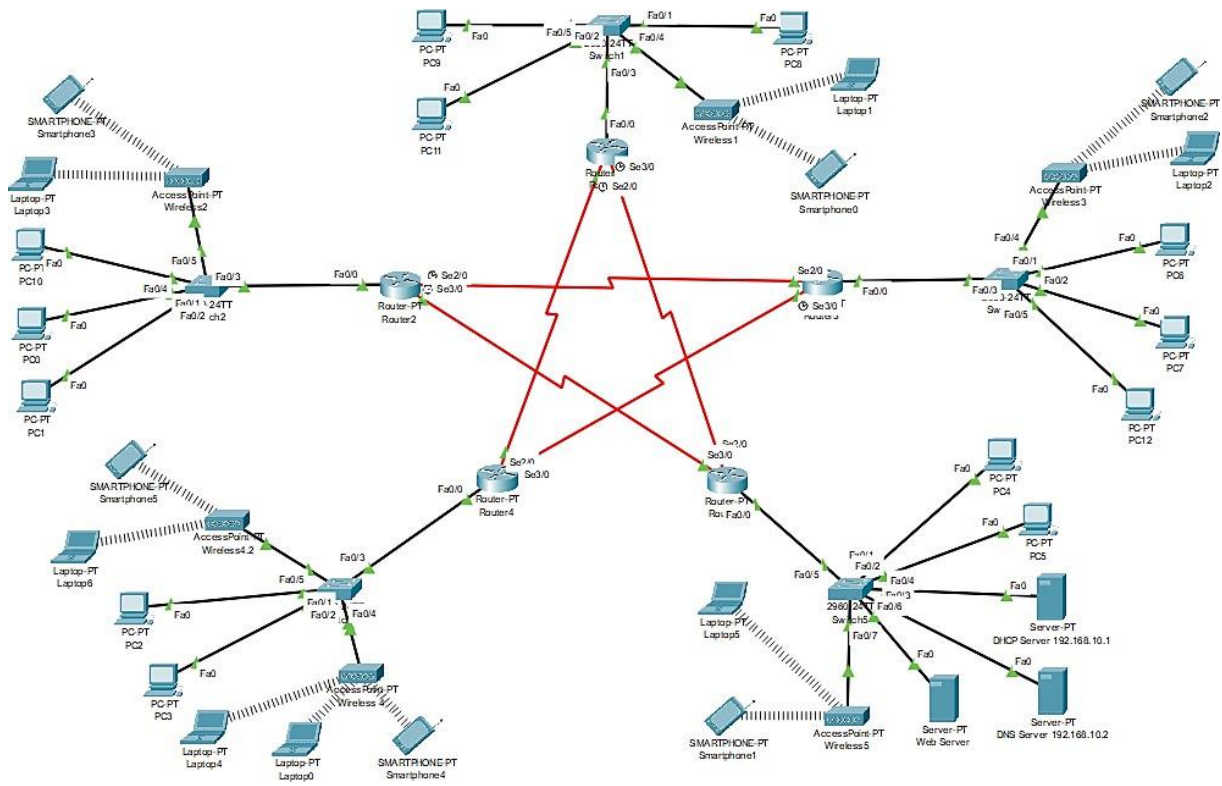


Fig-2

Router Setup

Routers 1-5 are for Network 1-5 in respective order.

Networks 1 and 2 are assigned to class A network where IP 90.0.0.0 is for network1, and IP 91.0.0.0 is for network2.

Networks 3 and 4 are assigned to class B network where IP 180.10.0.0 is for network3, and IP 180.20.0.0.0 is for network4.

Network5 is assigned to class C network where the network IP is 192.168.10.0.

Routers and intra-router networks are setup exactly as shown in Fig-1.

CLI commands used to setup each router is given below:

Router 1

```
interface fa0/0
ip address 90.0.0.254 255.0.0.0
no shut
do wr
exit
```

```
interface se2/0
ip address 192.168.30.1 255.255.255.0
clock rate 64000
no shut
do wr
exit
```

```
interface se3/0
ip address 192.168.20.1 255.255.255.0
clock rate 64000
no shut
do wr
exit
```


Router 2

```
interface fa0/0
ip address 91.0.0.254 255.0.0.0
no shut
do wr
exit
```

```
interface se2/0
ip address 192.168.40.1 255.255.255.0
clock rate 64000
no shut
do wr
exit
```

```
interface se3/0
ip address 192.168.50.1 255.255.255.0
clock rate 64000
no shut
do wr
exit
```

Router 3

```
interface fa0/0
ip address 180.10.0.254 255.255.0.0
no shut
do wr
exit
```

```
interface se2/0
ip address 192.168.40.2 255.255.255.0
no shut
do wr
exit
```

```
interface se3/0
ip address 192.168.60.1 255.255.255.0
clock rate 64000
no shut
do wr
exit
```

Router 4

```
interface fa0/0
ip address 180.20.0.254 255.255.0.0
no shut
do wr
exit
```

```
interface se2/0
ip address 192.168.30.2 255.255.255.0
no shut
do wr
exit
```

```
interface se3/0
ip address 192.168.60.2 255.255.255.0
no shut
do wr
exit
```

Router 5

```
interface fa0/0
ip address 192.168.10.254 255.255.255.0
no shut
do wr
exit
```

```
interface se2/0
ip address 192.168.20.2 255.255.255.0
no shut
do wr
exit
```

```
interface se3/0
ip address 192.168.50.2 255.255.255.0
no shut
do wr
exit
```

Routing tables were setup using OSPF dynamic routing algorithm.

Router 1

```
router ospf 1
network 90.0.0.0 0.255.255.255 area 1
network 192.168.30.0 0.0.0.255 area 1
network 192.168.20.0 0.0.0.255 area 1
exit
```

Router 2

```
router ospf 2
network 91.0.0.0 0.255.255.255 area 1
network 192.168.40.0 0.0.0.255 area 1
network 192.168.50.0 0.0.0.255 area 1
exit
```

Router 3

```
router ospf 3
network 180.10.0.0 0.0.255.255 area 1
network 192.168.40.0 0.0.0.255 area 1
network 192.168.60.0 0.0.0.255 area 1
exit
```

Router 4

```
router ospf 4
network 180.20.0.0 0.0.255.255 area 1
network 192.168.30.0 0.0.0.255 area 1
network 192.168.60.0 0.0.0.255 area 1
exit
```

Router 5

```
router ospf 5
```

```
network 192.168.10.0 0.0.0.255 area 1
```

```
network 192.168.60.0 0.0.0.255 area 1
```

```
network 192.168.50.0 0.0.0.255 area 1
```

```
exit
```

DHCP server is located at network5 and is tasked with automatically assigning IP addresses to hosts. Similar IP helper is commanded onto every network.

Command:

```
interface fa0/0
```

```
ip helper-address 192.168.10.1
```

```
exit
```

DHCP Server Setup

DHCP Server 192.168.10.1

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DHCP

Interface

FastEthernet0

Service

☒ On

☐ Off

Pool Name

serverPool

Default Gateway

192.168.10.254

DNS Server

192.168.10.2

Start IP Address :

192

168

10

10

Subnet Mask:

255

255

255

0

Maximum Number of Users :

246

TFTP Server:

0.0.0.0

WLC Address:

0.0.0.0

Add

Save

Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server
serverPool	192.168.10.254	192.168.10.2	192.168.10.10	255.255.255.0	246	0.0.0.0
serverPool4	180.20.0.254	192.168.10.2	180.20.0.10	255.255.0.0	512	0.0.0.0
serverPool3	180.10.0.254	192.168.10.2	180.10.0.10	255.255.0.0	512	0.0.0.0
serverPool2	91.0.0.254	192.168.10.2	91.0.0.10	255.0.0.0	512	0.0.0.0
serverPool1	90.0.0.254	192.168.10.2	90.0.0.10	255.0.0.0	512	0.0.0.0

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☐ Top

DNS Server Setup

DNS Server 192.168.10.2

Physical

Config

Services

Desktop

Programming

Attributes

SERVICES

HTTP

DHCP

DHCPv6

TFTP

DNS

SYSLOG

AAA

NTP

EMAIL

FTP

IoT

VM Management

Radius EAP

DNS

DNS Service

On

Off

Resource Records

NameType

A Record

Address

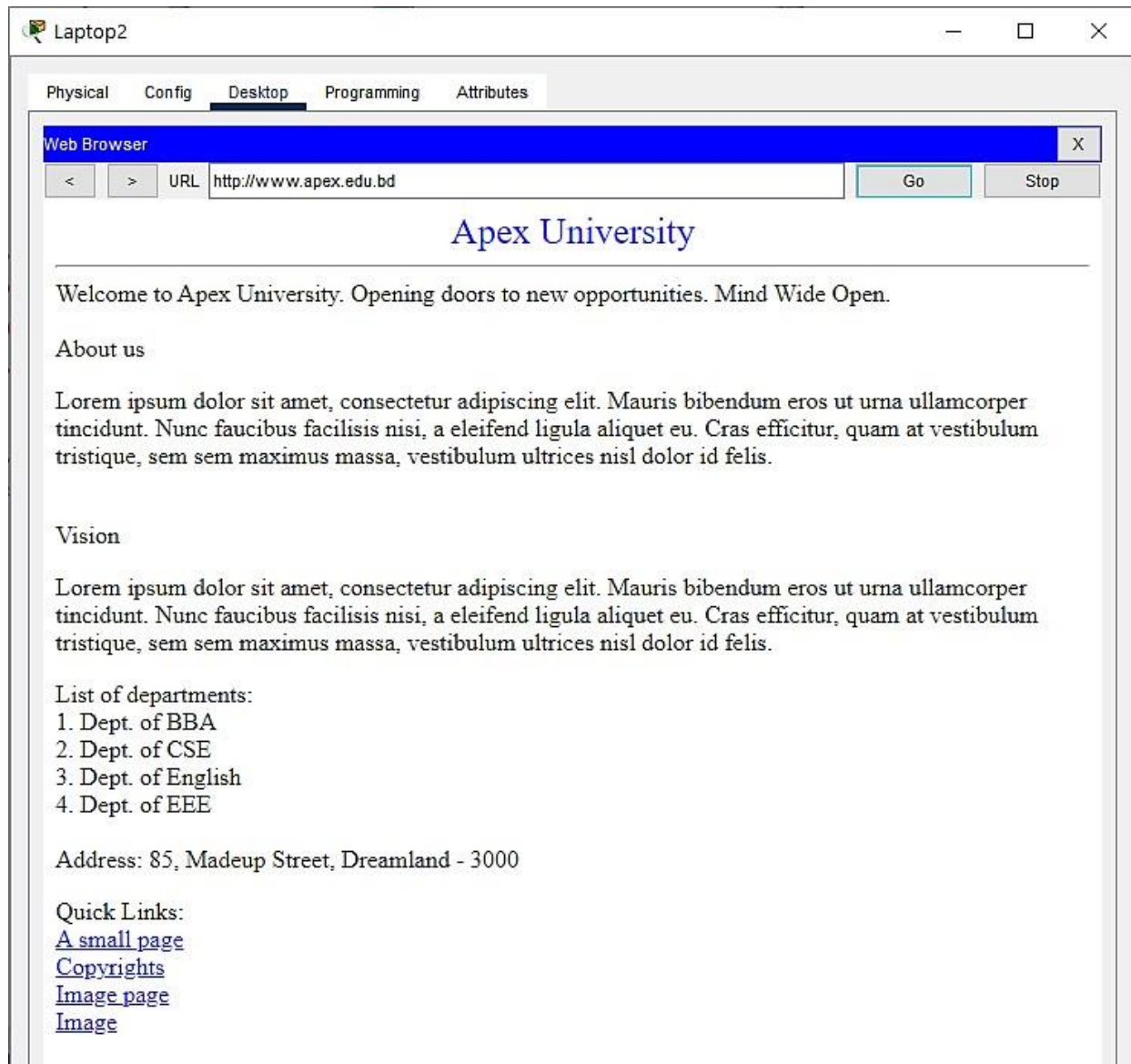
Add

Save

Remove

No.	Name	Type	Detail
0	www.apex.edu.bd	A Record	192.168.10.15

Web Page



Conclusion

In conclusion, university network systems play a crucial role in the functioning and operation of modern educational institutions. These systems can provide students, faculty, and staff with access to essential information and resources, such as class schedules, online course materials, and email. They also enable communication and collaboration among members of the university community, as well as facilitate administrative tasks such as grading and record-keeping. Overall, a well-designed and properly maintained university network system is essential for the success and efficiency of any educational institution.